

Securities Education Certificate



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Lecture 2

Macroeconomics + FX

Lecturers

Himanish Joshi

himanish.joshi20@imperial.ac.uk

Ahmad Danesh

ahmad.danesh18@imperial.ac.uk

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Imperial College Investment Society

Lead Lecturer

Ahmad Danesh

ahmad.danesh18@imperial.ac.uk

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Macroeconomics

Notes by **Himanish Joshi**

Lecturer, Securities Education Certificate 2022/23.

Sector Head (Financials) Queen's Tower Capital.

Introduction

The field of economics is divided into 2 sections, namely microeconomics and macroeconomics.

Microeconomics considers economics at an individual and business level, looking at decisions made that affect markets for singular products or services. Its areas of focus include resource allocation, price mechanisms, market failure and government intervention.

Macroeconomics however considers the economy at a country and international level, looking at policy decisions made by governments and central banks. Typical topics involve: inflation, interest rates, economic growth and employment.

Overall, microeconomics takes a “bottom-up” approach to analysing the economy whilst macroeconomics takes a “top-down” approach. Both areas are incredibly important in the field of finance, from private equity to global markets, from accounting to asset-management. In many ways, economics underpins many financial practices. To understand the rationale behind macroeconomics we will introduce some basic microeconomic theory such as supply and demand.

Microeconomics

Types of Economies

Fundamental concepts

Before we look at different types of economy, we must first introduce some very fundamental concepts. They are so fundamental in fact it can be summarised as the “Fundamental Economic Problem”. This is the idea that everyone has unlimited wants but only limited resources which is called scarcity. Unlimited wants refer to there being theoretically no end to the quantity or quality of goods that people would want to consume. However, there is only a finite supply of resources (raw materials) to fulfil those wants. This leads to choices. Decisions must be made: What goods should be produced? How should they be produced? How many should be produced? Who should consume them? Which allocation of resources is the most efficient? These decisions are economic decisions.

Another key concept closely related to scarcity is that of opportunity cost. Technically this is “the value of the next best alternative foregone”. Essentially this means the value of a choice that is not taken due to another choice being taken instead (a trade-off). Note that often, opportunity cost $\neq$ financial value. For example, by taking time to read these notes your opportunity cost is the time spent on something else you could be doing e.g. playing a sport, meeting friends or sleeping!

Now for the big questions: Who makes the decisions? Who determines the opportunity costs? You may think about a wide variety of groups: consumers, firms and governments... and you would be right! (but not all the time). The answer is: it depends on the type of economy.

Free Market

In a free market economy (sometimes called market economy), supply and demand (market forces) guide the allocation of resources. These forces determine the quantity and price of goods and services that are produced and consumed. Private businesses make economic decisions with minimal government intervention. This of course assumes that the market forces are the best mechanism through which to allocate resources in the economy. Theoretically, in a true-free market, there is no government intervention. In practice however, there is always some involvement by the government, meaning there are no real-world examples of true 100% free-market economies.

So, how do these “market-forces” lead to a theoretically efficient allocation of resources? It's through price discovery, which we will discuss in the next section. Essentially, buyers and sellers through supply and demand reach a price for every good/service known as equilibrium. Consumers drive choices through their decisions on goods and services – they influence the price set. Also, producers need to find a price by balancing their profit with customer affordability. The minimum government intervention allows competition in the market, which leads to efficiency. If one producer is charging too much for a product, another firm can enter the market and undercut them, driving the equilibrium price in the market lower. The competition is driven by a profit motive where businesses aim to engage in practices that increase profits by reducing costs and increasing revenues whilst remaining competitive in the market. Knock-on effects of competition include increased productivity and innovation.

However, there are drawbacks to free markets. For example, market failures can occur when supply and demand create prices that fail to reflect the social cost or social benefit of producing/consuming the product. Petrol/Diesel production is often quoted as a “negative externality” market failure. The argument is that the price consumers pay in a free market does not account for the cost of damages to the environment etc. and leads to a misallocation of resources. This leads to governments having to intervene through imposing fuel taxes to alter the price to a more “socially optimal equilibrium”. Another disadvantage of free markets relates to risks of the profit motive. To increase their profits, producers may cut costs through unethical means such as reduced health and safety for workers or damaging environmental practices. Finally, some economists believe that free-market practices lead to inequality.

Command Market

Completely opposite to the free market is a command economy. This is where resources are owned by the state or central government agency, and they decide how to allocate these resources without any market forces. The government sets production targets, salaries and growth rates according to its own view of people's wants. Advocates of command economies suggest that the central planning maximises social welfare and resources are allocated to reduce inequality and unemployment. This comes at the cost of less competition and innovation in the economy.

Most economies are mixed economies i.e. they share characteristics from free-market and command economies. Often, they allow market forces to determine the allocation of resources but do have government intervention to correct market failures. Some resources are owned by the public sector and some owned by the private sector.

Supply and Demand

Demand

How much do consumers want a good/service and their willingness to pay a price for them is encompassed in the idea of *demand*. More specifically, demand is the quantity of a good/service that consumers are *willing and able* to buy at a range of given prices during a given time period. Importantly “willing and able”, we are only interested in “effective demand” i.e. we only care about consumers who want something but are supported by an ability to pay for that thing. If you can’t pay for it, you have no effective demand. Throughout these notes, when we refer to demand, it will implicitly mean effective demand.

To analyse markets, supply and demand curves are drawn to better represent market prices and characteristics. However prior to drawing these curves we need to develop some theory into why they are shaped as such.

The **price** is what a buyer pays for one unit of a specific good/service. The number of units purchased at this price is the **quantity demanded**. The law of demand describes an inverse relationship between price and quantity demanded. For every extra unit demanded by a consumer they will demand a lower per-unit price.

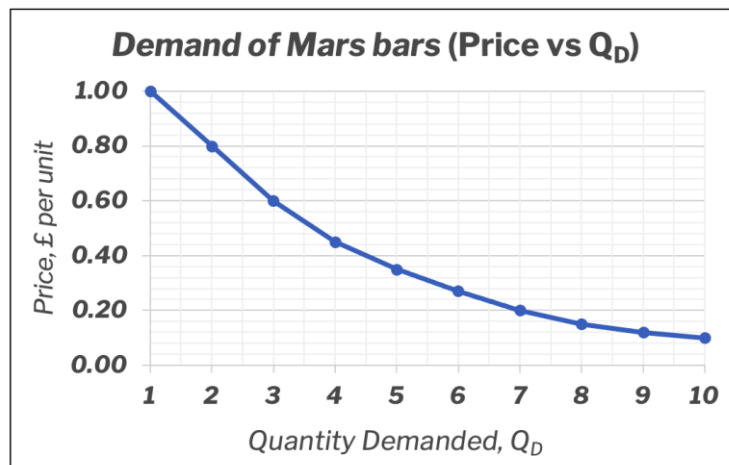
There are 3 main reasons as to why this is the case.

- 1) **Diminishing Marginal Utility:** The consumer gains less additional utility from consuming a product the more of it is consumed. For every extra unit, a consumer is willing to pay less due to the less utility (personal gain) they receive.

To illustrate this point, consider your demand for a Mars bar - assuming you like chocolate. Suppose there was 1 Mars bar and you would pay £1 for it. Now, consider there are 2 Mars bars. It's likely that the first bar will give you the most utility (i.e. pleasure). The second bar will also give you some utility but probably less than the first bar. This is due to the diminishing marginal utility. Would you still demand the 2 bars at a per-unit price of £1, knowing that the first bar was more “valuable” to you from a utility perspective? Rationally speaking, no. For example you may offer £0.8 per unit for the 2 bars. Now consider your demand for 10 bars. It's extremely unlikely that the 8th, 9th or 10th bar will give you much utility at all. Hence, you might rationally demand £0.2 per unit for the 10 bars.

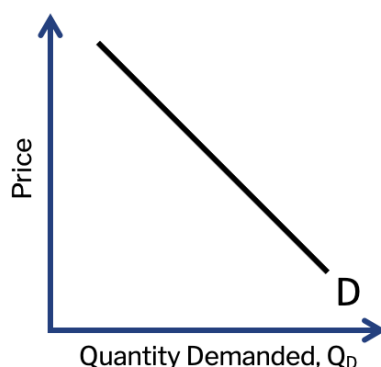
- 2) **Income Effect:** Consumers have fixed budgets to pay for the goods/services they consume. As the prices of goods increase, consumers can buy fewer units of the goods with the same income.
- 3) **Substitution Effect:** If goods become too expensive, consumers will find alternatives for it.

The aforementioned reasons lead to a “demand schedule” - a table which illustrates the law of demand. From this we can plot a demand curve with price on the y-axis and quantity on the x-axis:



Price (£ per unit)	Quantity Demanded (units)
1	1
0.8	2
0.6	3
0.45	4
0.35	5
0.27	6
0.2	7
0.15	8
0.12	9
0.1	10

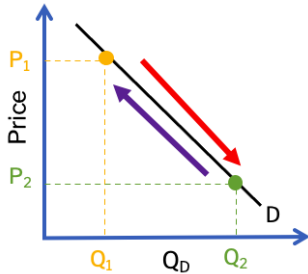
Demand curves can look different for various goods/services as they may have different gradients and curvatures. However, they all exhibit the fundamental property of sloping down from left to right due to the law of demand: as price increases, the quantity demanded decreases. We can simplify all demand curves as a linearly downward sloping curve. The gradient of the demand curve has an interesting property called “elasticity of demand” that will not be explained here but those interested should read into this. N.B. sometimes quantity can be referred to as ‘output’.



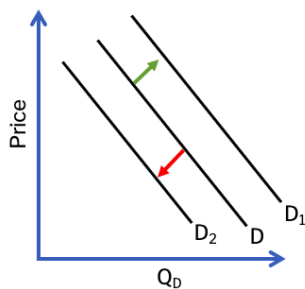
A very niche exception to the downward sloping demand curve is the notion of “snob goods” which lead to an upward sloping demand curve. These occur when the goods are valued highly purely because their price is high and consumers want other people to see that they own these goods. Consider luxury items such as

jewellery, Rolex watches, yachts etc. This concept of “snob goods” only applies to these luxury goods and will be ignored in future references to the concept of demand.

So, demand curves seem useful but their true power is reflected when we consider changes of the curve. There are 2 types of changes to consider: movements along the curve and shifts of the curve.



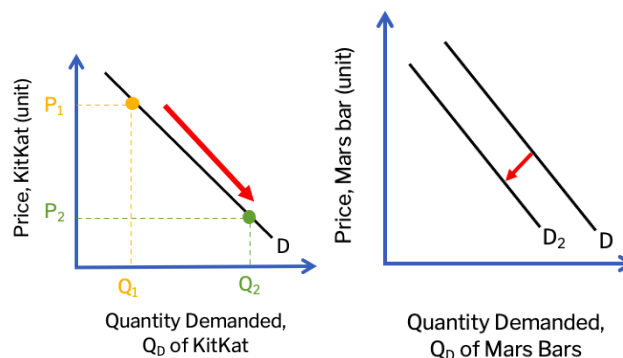
A movement along a demand curve can only be caused by a change in the price. If the price decreases from P_1 to P_2 this is accompanied by a rise in quantity from Q_1 to Q_2 . This movement is the red arrow and is described as an “extension in demand”. Conversely a “contraction in demand” (the purple arrow) shows a situation where price has increased, and output has decreased.



A shift of the demand curve is more interesting. It occurs when the quantity demanded changes for every price along the curve, which happens when any factor other than price changes the demand for a good/service. The green arrow shows an outward shift of the curve which is an increase in demand. The red arrow shows an inward shift of the curve representing a decrease in demand.

You may be wondering what would cause a shift in demand. Many reasons in fact can result in changes in demand.

Firstly, changes in income. If people have more money to spend, they will likely be “willing and able” to demand a good/service at higher prices thus resulting in a shift out of demand. An increase in population can result in increased demand. Another interesting cause is a change in the price of substitute goods. If the price of KitKats is reduced, then by the law of demand consumers will demand more KitKats. Therefore, they have less money to spend on Mars bars thus resulting in a shift inward (decrease) in the demand of Mars. The following diagrams illustrate this scenario.



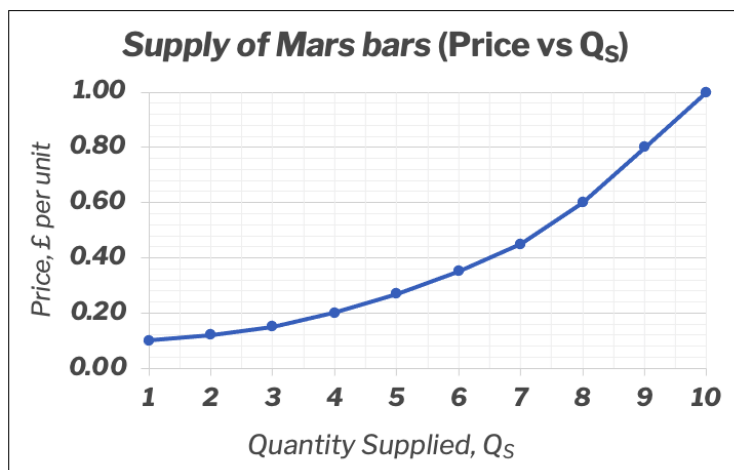
You may be wondering how analysing these microeconomic supply and demand diagrams will aid us in understanding global macro. As we will see later on, macro diagrams will be very similar and follow the exact same principles!

Supply

Now that you’ve understood the concept of demand, we will look at supply.

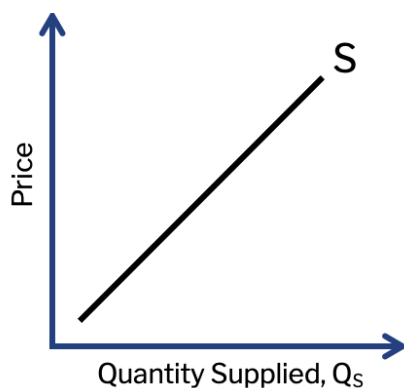
The law of demand was largely driven by consumers. Supply, however, is explained by looking at producers (i.e. firms). Supply is the quantity of a good/service that firms are willing and able to produce at a range of prices in a given period. In free-market economies, firms are driven by a profit motive. They make more profit as prices of the goods/services they sell rise. This suggests that firms are willing to supply a greater quantity of a good as its price increases thus causing a positive relationship (upward sloping curve) between price and quantity supplied. This is known as the law of supply.

Going back to the example of Mars bars. Suppose that the price of the bars increased, well then its rational to assume that Mars will want to produce more bars, to take advantage of higher prices and make more profits. This may lead to the following Supply schedule and curve:

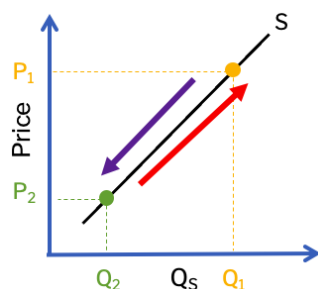


Price (£ per unit)	Quantity Supplied (units)
0.1	1
0.12	2
0.15	3
0.2	4
0.27	5
0.35	6
0.45	7
0.6	8
0.9	9
1.0	10

Supply curves also have different shapes but all exhibit the fundamental property of the law of supply, a positive relationship between price and quantity. We can simplify such supply curves as a linear straight line with positive gradient. As with demand, the gradient of the curve is called the “elasticity of supply” and has properties that relate to the type of good/service.

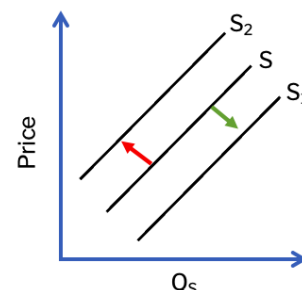


Like demand curves, the supply curves can also change through movements along the curve or shifts of the whole curve.



A movement along a supply curve can only be caused by a change in the price. If price decreases from P_1 to P_2 (left) this is accompanied by a decrease in quantity from Q_1 to Q_2 . This movement is the purple arrow (left) and is described as a “contraction in supply”. Conversely, an “extension in supply” (the red arrow, left) shows a situation where price has increased, and output has increased.

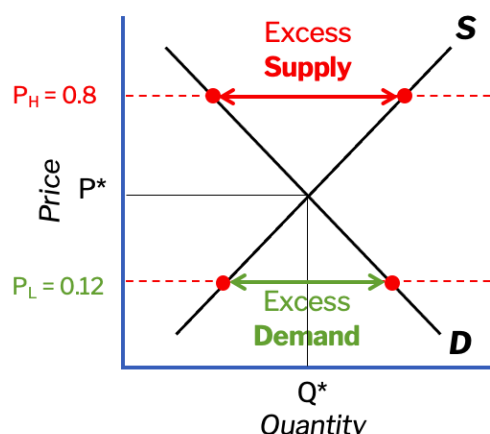
A shift of the supply curve occurs when the quantity supplied changes for every price along the curve, which happens when any factor *other than price* changes the supply for a good/service. The green arrow (right) shows an outward shift of the curve which is an *increase* in supply. The red arrow (right) shows an inward shift of the curve representing a *decrease* in supply.



Factors that cause shifts in supply curves are largely related to what determines a firm's profits. Anything that increases a firm's costs of production will reduce supply as the profit per unit decreases and firms will be less willing to produce the good/service. For instance, a rise in rents, wages, raw material costs etc. will all increase costs of production and lead to a decrease of supply (shift from S to S_2). Changes in technology can help increase supply e.g. new machinery that makes production lines more efficient and reduces production costs per unit. Government policy can also shift supply curves via taxes and subsidies. Subsidies given to solar panel manufacturers help to reduce the firm's costs leading to an outward increase in supply (shift from S to S_1). On the other hand, taxes on the production of goods lead to reductions in supply as they increase costs for firms. For example, carbon taxes on oil production are designed to encourage firms to reduce their output.

Market Equilibrium

Now that we have seen both demand and supply curves, it makes sense to plot them together. By doing so we can visually show the market of any good/service.



Market equilibrium is defined as the price and quantity at which the quantity demanded by consumers is exactly balanced by the quantity that firms wish to supply.

On the graph above, this is shown as the intersection of the supply and demand curves. P^* and Q^* are the symbols used to depict the equilibrium price and quantity. You may be wondering ‘why is it that the market equilibrium must be the intersection?’ Well, to explain this, we can consider what happens when the market is not at equilibrium.

Assume this is the market for Mars bars as shown earlier. If we were operating at P_H (£0.8 per unit) then the quantity demanded by consumers is 2 units. However, the suppliers are willing to supply 9 units. There are 7 units of excess supply in the market which is an inefficiency. The supplier will be faced with depreciation costs and cannot make any profit from the extra 7 units they hold. Market forces, supply and demand, dictate that the price must be lowered to reach a more optimum output for both consumers and firms.

Conversely, if the market operated at P_L (£0.12 per unit) then the quantity demanded is 9 units, but the quantity supplied is only 2 units. This time, we have excess demand of 2 units. There is once again an inefficiency in the market that can be corrected by raising price and output, enabling more consumers to consume the goods and firms to make more profits via higher prices. This convergence of price and quantity to the equilibrium is the price mechanism ("market forces") by which free-markets aim to operate efficiently. Note that for this example, it turns out that P^* is around £0.30 and Q^* is between 5 and 6 units.

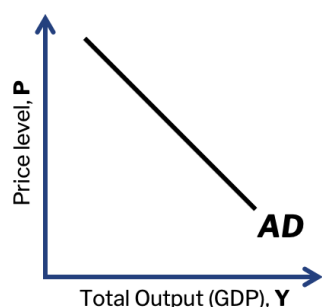
Let's now transition from the microeconomics world to the macroeconomics world.

Macroeconomics

Aggregate Demand (AD)

Aggregate Demand (AD) is the total amount of spending on goods and services produced in an economy during a period of time. It can be thought of as Gross Domestic Product (GDP) which is a measure of the market value of all the final goods and services produced and sold by a country in a specific period.

In microeconomics we considered the *demand* for one good/service. Here we are just scaling the concept up: instead of one good/service, we consider *all* the goods/services in the economy. Once again, the law of demand applies, and the AD curve is shaped *exactly* like an ordinary demand curve. This time the x-axis represents total output, and we use the symbol Y . Total output is also correlated with the level of employment in the economy. The y-axis represents price-level and loosely speaking "inflation" and we use the symbol P .



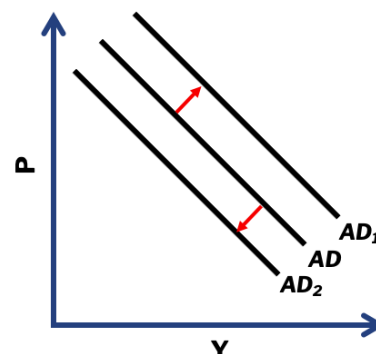
AD can be broken down into separate components: consumption, investment, government spending and net trade. We can express this conveniently as:

$$AD = C + I + G + (X - M)$$

where X represents exports and M represents imports. A change in any one of these components will result in a shift of the AD curve either outwards (increasing AD) or inwards (decreasing AD).

The largest component of AD is Consumption, C (which typically makes up around 70% of total AD). Consumption largely depends on disposable income. Higher levels of disposable income led to greater consumption as individuals can afford to buy higher quantities of goods/services at higher prices. This would lead to a shift out of AD to AD_1 .

Another important factor affecting consumption is interest rates. In macroeconomics, interest rates refer to the cost of borrowing i.e. how



expensive it is to take out a loan or a mortgage. It also refers to saving i.e. how high are the rewards for saving money? Interest rates are managed by central banks and are crucial to managing the economy and will be explored later when we discuss monetary policy.

Theoretically, interest rates can be thought of as the opportunity cost of holding money. If you store money under your bed (which you really shouldn't), you are forgoing the interest rate of the return on any other asset, noting that return can be negative or positive. Anyway, the important thing to note here is that interest rates should be thought of as the cost of borrowing. So, high interest rates would mean a higher cost of borrowing, meaning that credit becomes expensive and saving more attractive leading to a decrease in consumption and a decrease in AD to AD_2 .

Investment refers to capital expenditure undertaken by firms to add to their capital inventory (machinery, factories or new technologies). Government spending largely depends on the condition of the economy (*business cycle*). During times of recession e.g. during the Covid lockdowns, the government was forced to increase its welfare spending on unemployment benefits (e.g. furlough), grants and bailouts. In addition to welfare spending, there is public sector service spending e.g. increase in funds to the health service. These types of spending programs by the government aren't entirely funded by taxes. Often, it would involve monetary policy such as quantitative easing (which we'll explain later). Finally, net trade affects AD. This depends on numerous factors including exchange rates, the state of the world economy, degrees of protectionism (tariffs etc.) and consumer demand.

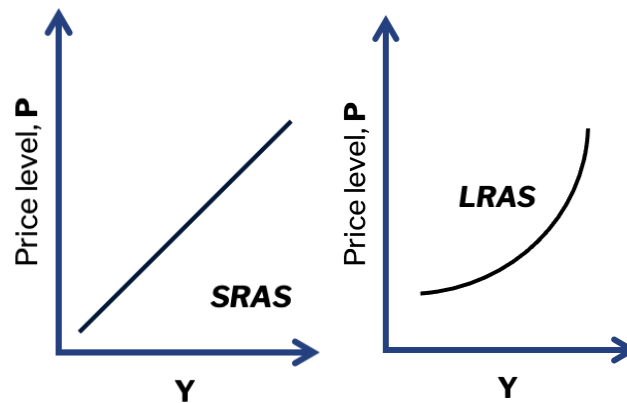
Aggregate Supply (AS)

We can extrapolate the *supply* we looked at earlier to encompass every good/service in the economy and we would get Aggregate Supply (AS). Remember that supply is looked at from a firm perspective **NOT** the consumer. So, AS is the total value of output that firms are willing and able to supply at a given point in time.

Unlike AD, there are 2 types of AS curves: SRAS (Short Run Aggregate Supply) and LRAS (Long Run Aggregate Supply). This is because the ability of firms to change production levels differs when we consider different timeframes.

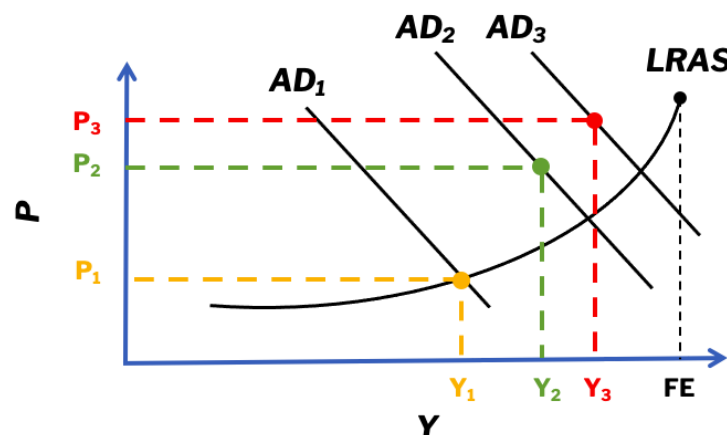
Suppose you own a car firm. In the short-run, your ability to change the number of cars you produce would largely depend on the costs of production. If energy prices increased overnight, it is likely your supply may slightly decrease. On the other hand, if energy prices went down it is unlikely that you would be able to increase your production by a meaningful quantity. Hence for SRAS, the ability for firms to vary output is limited and SRAS can only shift due to a change in cost of production. Suppose energy prices went down and you used that excess capital to invest in a new technology. Over the long run, this may lead to your factory being more efficient and the quantity of cars you produce would increase noticeably. Hence for LRAS, the ability of firms to vary output is flexible. This difference leads to 2 differently shaped AS curves that both still follow the law of supply.

Interestingly, there are multiple types of LRAS curves and the interested reader should Google about Classical and Keynesian views of AS. As macroeconomic policy is largely associated with the long-run, we will mainly be considering the LRAS. Once again, the y-axis represents the price level and the x-axis represents total output/employment.



Shifts in LRAS are caused by anything that increases the quality or quantity of resources used to produce goods/services. Technological advances can increase productivity of supply chains and help decrease the cost per unit whilst also attracting new firms into the market. Education and skills improvements can make workers more efficient and increase the quality of the labour force thus enabling more goods/services to be produced and increasing the AS in the economy. Finally (and perhaps most importantly), changes in government policy. This can include fiscal policy such as taxes and welfare spending or monetary policy such as interest rates and financial market interventions.

To help explain how price level (inflation), output and employment are all related we will consider a series of situations.



Here we are representing changing macroeconomic environments through increasing AD curves. FE denotes *full employment* and it is the maximum possible output of the economy if all resources are used and the LRAS is constant.

Consider AD_1 first. In this situation, the equilibrium point creates an output of Y_1 and price level of P_1 . We can see that we are operating far from full employment. This means there is a lot of *spare capacity* in the economy. When I mention employment, I refer to the fact that more resources can be used, more people can be hired and more goods can be produced – **NOT** just employment in terms of people and job opportunities.

As we shift to AD_2 we move closer to FE and some of the unused spare capacity has been used up, which has enabled an increase in output and in employment. At the same time, we see the price level of the economy increasing from P_1 to P_2 . Why is that? Well those extra employees need to be paid and to gather the extra resources costs money. Overall, *costs of production for firms have increased*. This is inflation. The increase in price however is minimal and this is overall healthy for the economy.

Now suppose AD continues to shift to AD_3 . Now we are practically at full employment and there is no spare capacity left. The cost of acquiring additional resources has gone up exponentially. As there is so little extra resource, firms must pay much more to acquire it. This translates to higher costs of production and much greater prices in the economy. This has led to very high inflation as price levels have risen from P_2 to P_3 . Now that we have reached peak output of the economy you may be wondering how is it possible that economies continue growing? Well this is achieved through increasing (shifting out) the LRAS (notice that in this example, we assumed LRAS was fixed). We have seen earlier that very high inflation can be disastrous for the economy and in many ways the point P_3 is indicative of global economies today. Government intervention is required, and this will be discussed in the next section.

Inflation

If you were not sleeping under a rock, you may have noticed that the price of almost everything has increased. Food, fuel, transportation, entertainment and rent have all significantly risen in price. Undoubtedly, you must have felt that a pound today does not stretch as far as a few years ago. This is **inflation** – the sustained rise in general price level over time.

Very high rates of inflation do indeed “make everyone poorer” as the cost-of-living increases and the purchasing power of money decreases. Another negative consequence of high inflation is that it disrupts business planning by discouraging capital investment and degrading business confidence. One of the most important consequences of inflation and what is extremely topical is the concept of a wage-price spiral. As inflation rises (currently between 8-10% since August in the UK), workers demand higher wages. For example, transport staff have been striking in demand for higher wages. If employers do indeed decide to increase wages, this can lead to inflation spiralling out of control as higher incomes will further increase consumption and demand in the economy putting further pressure on prices (a shift out of AD)

However, some inflation is a good thing. To illustrate this, we can consider the opposite of inflation: **deflation**, which is a decrease in price level over time. The most notable example of this is Japan, which has had decades of extremely low or negative rates of inflation. If prices of goods/services were continuously decreasing, you may think this is good right? Everything is getting cheaper! But this is disastrous for the economy, as you as a consumer are likely to delay your spending choices.

For example, that TV you were thinking of buying. Well... why not wait a couple of months and you could get it at a lower price? This can be extrapolated to firms wanting to make investment decisions e.g. building new factories or researching new technologies. Delayed spending decisions decrease the aggregate demand in the economy and reduce overall output (i.e. GDP). Firms are less likely to invest into new technologies and the economy stagnates. Overall, low and stable inflation is a positive for the economy. In the UK, the target inflation rate set by the government is 2% and it is managed by the Bank of England. Low and stable inflation is often a consequence of sustainable growth and a healthy macroeconomic environment.

The most quoted measure of inflation is the % change in CPI (Consumer Price Index). CPI measures price levels in the economy by considering the price changes of a basket of goods/services. What goods are in the basket are determined by a national agency, in the UK this is the ONS (Office for National Statistics). A slightly more detailed explanation of the calculation is below:

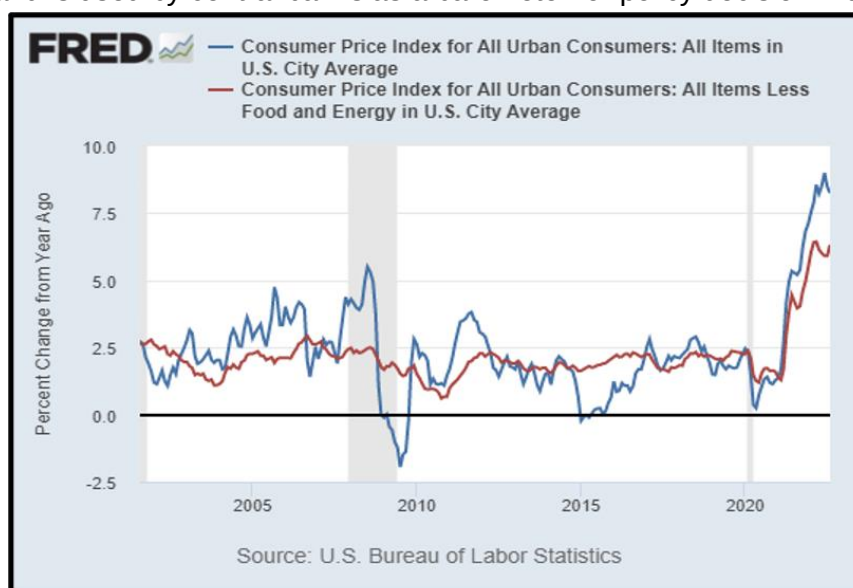
- 1) A base year is chosen where the average price of the basket is measured, and a corresponding index value is set to 100.
- 2) Each subsequent quarter or year, the changes of prices of goods in the basket is determined and a new index value is calculated. We can estimate inflation by looking at the % changes of the index value from the base value of 100.

- 3) The goods/services in the basket are weighted to reflect the relative importance of that good to a typical household. Around 700 goods are used to calculate the CPI.

You may be wondering that it is subjective which goods should be in the basket and how they should be weighted. You are correct! One of the limitations of the CPI is that it is difficult to define typical consumption (since different households can have drastically different spending patterns). Another drawback of CPI is that it doesn't include mortgage interest payments (the biggest monthly expenditures of many households) and that the CPI is slow to react to new inclusions in the basket.

To resolve this, there is another measure of price-levels called RPI (Retail Price Index) which does include mortgage payments and is based on a different basket of goods. Furthermore, the RPI covers a smaller range of households (e.g. it excludes pensioners/high income earners), meaning it aims to target the average household. Having said this, the CPI is a more useful indicator to measure policy effectiveness due to its tried and tested calculation methods and its worldwide adoption making it helpful in performing international comparisons.

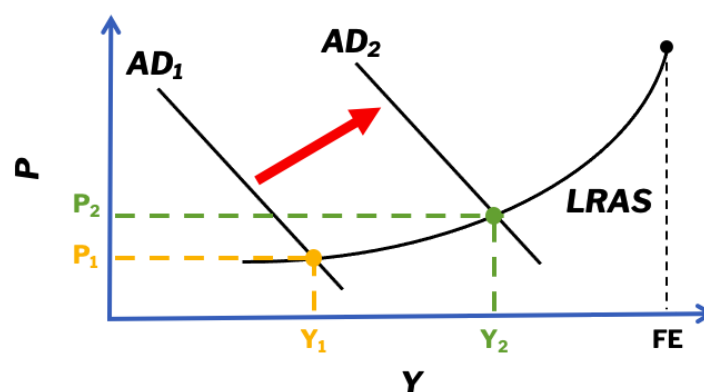
A very important extension of CPI is Core CPI. The normal CPI reading (often called 'headline inflation') includes all parts of the economy which experience inflation and is not adjusted to remove volatile readings that are not caused by changes in economic conditions i.e. food and energy prices which can exhibit extensive volatility due to climate variances, political turmoil, natural disasters etc. Core CPI removes these volatile components (food and energy) from the basket of goods/services. This is deemed as a better measure of economic conditions and is used by central banks as a barometer for policy decision-making.



The graph above shows CPI inflation (blue) and Core CPI inflation (red) in the US over the past 20 years. We can see that the headline reading (which includes food and energy) is much more volatile than the core reading. Other key things to note include the huge fall in inflation during *and* after the global financial crisis from 2007-2009. So much so that inflation even went negative (i.e. prices were falling!), but as discussed this is a bad thing.

We can use the aggregate supply and demand diagrams to illustrate the causes of inflation. The first type is **demand-pull inflation** where AD has shifted out raising prices and output of the economy.

Demand-pull inflation



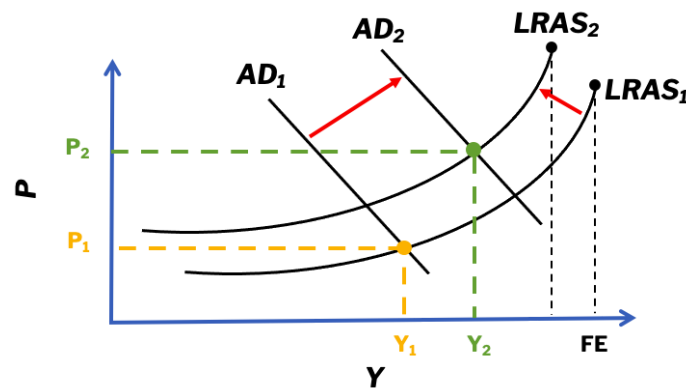
In the above diagram, we can see that AD has shifted outward (increased) from AD_1 to AD_2 . Remember that AD shifts when anyone of its components in the equation $C+I+G+(X-M)$ changes. The diagram above is a good example of the recovery after the Covid lockdowns. When restaurants, entertainment and travel opened again, there were big increases in consumer spending (consumption) for these goods/services. Furthermore, as consumption is the largest component of AD, it is likely that there would be a significant increase in AD. At the same time when we had an increase in consumption, we also had increases in government spending. The government was forced to increase welfare payments (through furloughs, unemployment benefits etc.) to ensure people could survive despite not going to work. This excessive government spending creates more credit (i.e. “money”) in the economy and further increases AD.

Cost-push inflation

The second cause of inflation is called cost-push inflation. Here, the root cause is on the supply-side and more specifically, decreasing supply. This is often due to rising costs of production. This might be indicative of the current situation where the war in Eastern Europe has led to surging commodity prices and raw material costs for firms. As these costs increase, firms are restricted in the quantity of goods/service they can provide and hence AS decreases. Rising energy prices are **NOT** the only reason that AS has been decreasing. Supply side problems are present across the world due to many reasons such as lockdowns in Asia, the inability of firms to find workers in the transportation/shipping industry in the US and rising interest rates.

You may be wondering ‘aren’t we currently experiencing scenario one (demand-pull) and scenario two (cost-push) inflation?’ Yes, we are! We can plot both these on the same diagram to truly understand the current macroeconomic environment.

Combining demand-pull and cost-push inflation



The above diagram is a very good representation of many global economies. The important things to note are an increase in AD and a decrease in AS. This has caused a huge increase in prices as reflected by globally high inflation levels with a moderate increase in output and employment. This makes sense as the economies are recovering from the pandemic where GDP and employment fell tremendously. Currently the big problem is high inflation. To tackle this, we will look at government policy.

Fiscal and Monetary Policy

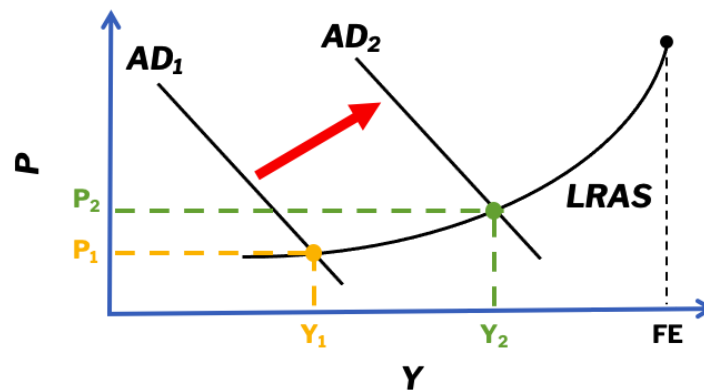
We have seen that AD and LRAS curves can be used to effectively portray macroeconomic environments. The market forces on their own dictate several important economic factors such as: GDP, employment and inflation. These factors in turn have huge knock-on effects to our everyday lives via the quantity/quality of goods we can consume, cost of living, quality of life and household incomes. If left to their own devices, market forces can sometimes create situations that are unfavourable to the economy (think high inflation or low employment etc.). So, how can economies artificially “control” AD and LRAS to ensure that situations like these (market failures) are minimised? The answer is Government Policy. More specifically, a government can intervene to influence the markets via fiscal policy and monetary policy.

Fiscal Policy

Fiscal policy is the manipulation of AD in the economy via changes in taxation or government spending. It can also have supply side effects and be used to improve education, health or the redistribution of income. To see this in action, we can consider the fiscal policy response of the UK government to the Covid pandemic. In March 2020, the government implemented several policies that included social distancing, shutting down of shops, travel restrictions etc. that led to a serious downturn in the UK economy (e.g. 19.5% drop in QoQ* GDP for Q2 2020 and an overall contraction of 9.8% in 2020). This was a situation where AD had severely decreased, people weren’t buying much so consumption decreased, businesses weren’t investing and overall the economy was contracting. Consequently, a policy needed to be adopted that increased AD.

We need to think of the components of AD i.e. $C+I+G+(X-M)$, remembering that consumption accounts for around 70% of total AD. Naively speaking, we need a policy that can get more money into the hands of people hoping that they will spend it and in turn increase economic activity (AD). The UK Government attempted to do this by increasing welfare payments (e.g. furlough payments at an estimated cost of £70bn, an increase in Universal Credit and grants to businesses). This policy aimed to increase the **C** and **G** components of AD. Also, the government pledged to spend £1bn on grants for firms committing to innovation, aiming to increase the **I** component. These measures (among others) are examples of fiscal policy trying to influence the market and boost AD.

*QoQ = Quarter-on-Quarter



You've already seen this graph so we can use this to explain the fiscal policy that the government tried to pursue during the pandemic. Note that fiscal policy has a time-lag, meaning there is a delay between when policy is introduced and when its effects can be seen in the market. Furthermore, there can be unintended consequences. For example, what if the extra credit in the economy was not spent on goods/services but was instead saved for future use? This concept is called the "multiplier" and interested readers should look up "marginal propensity to save / marginal propensity to consume".

In my opinion, the more interesting intervention is monetary policy. Whilst sometimes governments play a role in this, largely monetary policy is driven by central banks. In the UK this is the Bank of England (BoE), whilst in the U.S. it is the Federal Reserve (Fed) and in Europe it is the European Central Bank (ECB). The primary tool behind monetary policy is interest rates. As discussed earlier, interest rates should be thought of as the cost of borrowing.

But, how do interest rates affect AD and LRAS and consequently the overall economy? Well... in many ways! Firstly, we will consider the impact of decreasing interest rates on AD. Low interest rates encourage consumer borrowing – as the saying goes: "money is cheap". Every form of loan becomes cheaper, from car loans to bank loans to credit card loans. Overall, there is more credit in the economy and disposable incomes for households increase. In turn, consumption of goods/services increases. At the same time, saving becomes less attractive, further increasing consumption.

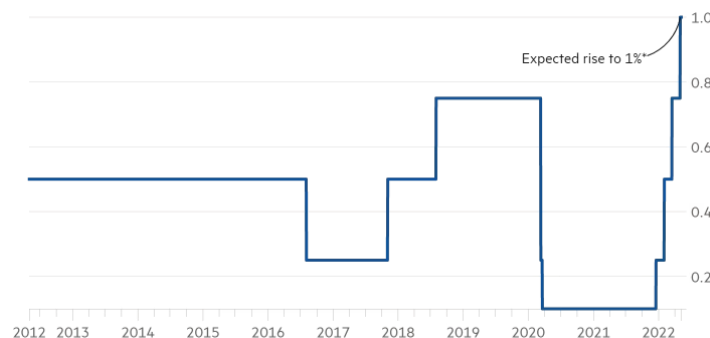
Simultaneously, as interest rates fall, mortgage payments decrease, and this can trigger housing booms as more people can afford the monthly payments and demand for housing increases. From a business perspective, borrowing costs decrease. This encourages investment as loans for big capital projects (new factories, research and development, mergers etc.) come at lower premiums.

Finally, usually (but not always) a fall in interest rates typically leads to a depreciation in a country's currency in FX markets. This means imports become more expensive whilst exports seem cheaper. Overall, interest rates have a wide-ranging effect on the economy, and this is what makes monetary policy so powerful. Unlike fiscal policy which largely has targeted effects, interest rates affect so many different groups of the economy through many different channels and can lead to a more coordinated response when intervening in the markets.

Interest rates also have supply side effects and can influence LRAS. For example, low and stable interest rates may encourage firms to invest through big capital projects as they can obtain relatively cheaper financing. In the long run, this can lead to improved efficiency and lower costs for firms meaning that LRAS can increase (shift outwards). Here, we explained the situation where interest rates were decreasing, and this typical policy undertaken during times of economic slowdown e.g. during the COVID pandemic.

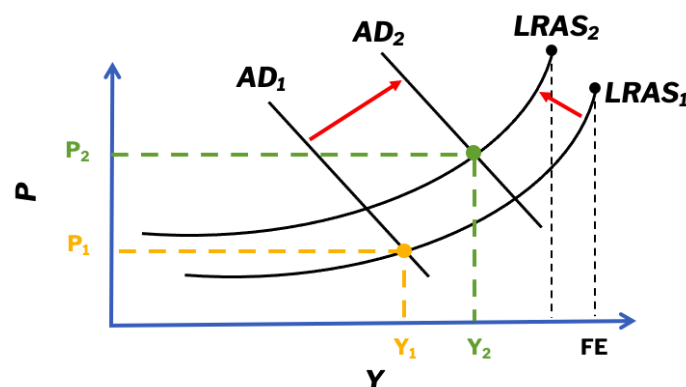
The Bank of England is expected to raise rates again this week

Bank of England rate (%)



*Based on Reuters poll of economists
Source: Bank of England
© FT

The graph above illustrates the drop in interest rates to historic lows of 0.1% by the BoE to try and boost economic activity during the pandemic 2020. Interestingly, there are a series of spikes from early 2022 this year where rates have suddenly shot up and are currently at 2.25%. Why have interest rates gone up so much and why so quickly? Well to answer this we can consult the earlier diagram we drew about the current macroeconomic conditions (see below).



The current macro environment is dominated by high inflation (8.6% CPI in the UK, 8.3% CPI in the US and 10% in the eurozone). This is due to the demand-pull and cost-push factors that were explained earlier. To reduce inflation, central banks have been raising (*hiking*) interest rates at unprecedented speed. This is a form of *contractionary* monetary policy where the aim is to tighten financial conditions by making money more expensive to borrow. In theory, this has the reverse effect of those outlined earlier leading to a decrease in AD and reducing inflationary pressures.

Another form of monetary policy is quantitative easing (QE). This is performed in times of economic downturn and typically occurs in conjunction with a decrease in interest rates. Here, a central bank starts buying up bonds issued by the government (which you'll cover in greater detail in lecture 6 with Don-Angelo). Do you remember the increased welfare spending we mentioned earlier? Well, think about it: how does the government obtain the money to execute such large spending programs? It is debt. More specifically, government bonds (rates – a fixed income instrument) which are issued by institutions in return for coupon payments and return of the principal at some maturity date in the future.

So in QE, central banks buy up these government bonds, in turn driving the price of the bonds higher. As the prices of these bonds increase, their yield (coupon payment as % of price) decreases. This "yield" can be thought of as an interest rate and is often the benchmark interest rate used on many loans. Sounds like earlier

now, right? These lower yields feed down through the economy and seek to decrease the cost of borrowing and hence increase spending/consumption in the economy. For perspective, as part of the response to the Covid pandemic the BoE adopted a QE program worth up to £450bn.

The reverse of QE is Quantitative Tightening (QT) where central banks reduce their bond holdings. Expectedly, this has the opposite effect of QE. This time, bond prices drop and yields rise, having the ripple effect of higher interest rates and reducing the amount of spending in the economy. Currently, in response to higher inflation and a need to reverse the extensive QE programs during Covid, many central banks have been forced to start QT and this has led to falling bond prices across the world.

In conclusion, by reading these notes you should have a good understanding of the mechanics behind demand and supply. Moreover, you should better understand macroeconomic concepts such as inflation, AD and LRAS. Finally, you will now appreciate the importance of interest rates and monetary policy in influencing macroeconomic environments.

Image Credits

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Image Source: FRED Federal Reserve Bank of St Louis

Data Source: US Bureau of Labor Statistics U.S. Bureau of Labor Statistics, Consumer Price Index for All Urban Consumers: All Items in U.S. City Average [CPIAUCSL], retrieved from FRED, Federal Reserve Bank of St. Louis; <https://fred.stlouisfed.org/series/CPIAUCSL>, October 10, 2022.

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Image Source: Financial Times, 2 May 2022, "Bank of England poised to raise interest rates further to curb inflation"

Data Source: Bank of England

Foreign Exchange

Notes by **Ahmad Danesh**

Lead Lecturer, Securities Education Certificate 2022/23.

Founder, QT Capital.

A basic introduction to FX

If you're sitting here thinking this has anything to do with Forex scams, you'd be completely right! Well, no. Some of the concepts you will be studying here are the same ones that those dodgy rental Lamborghini-driving, rental mansion-living, rental Gucci-wearing 25-year-old snake oil salesmen try to use in order to lure you in to exchanging your hard earned money for the false promise of a return based off some 'mArKeT sEnTiMeNt'. In the SEC, we'll be discussing the ways that Foreign Exchange (FX) can be used as a way for well-established market players including hedge funds and other institutional investors to hedge against changes in currency prices by analysing its basic features, linguistics and how to read things such as currency pairs. Why all of this is relevant will become clear in Lecture 4 where myself (Ahmad) and Oscar will be discussing the ideas of derivatives.

So, why would institutions care about foreign exchanges? First of all, what is a foreign exchange? When you hear the word 'foreign exchange', the first idea you'll need to be aware of is a *foreign exchange rate*. This is simply how much money you need to convert from one currency to another one. For instance, an exchange rate of £1 = \$1.21 means that to get a £1 coin using the dollars in your wallet, you need to cough up exactly \$1.21. The place where you do this exchange of currency is at a *foreign exchange market*. Cool, huh? This doesn't seem too trivial if you go on holiday. After all, if you travel from the USA to the UK, you won't be able to walk into a Tesco and pay for a bottle of vodka at £10 using a \$10 note, a \$2 and 10 cents – it's actually an enforced law that shops only accept Great British Pounds (GBP), which is why you walk into a Bureau de Change at Heathrow or wherever to swap your currency for the local one. Some people make the mistake of attempting to pay using the 'face' value of an item i.e. paying for a £10 product with \$10. You can see using the exchange rate why that concept doesn't work.

Now, why do institutions care about foreign exchanges? Well, often in this world of globalisation, a customer (whether that might be an individual, a company or an institution) might need to pay for a product that's not made in the country where the customer themselves are located. Apple don't make the electronics for the iPhone in the US – it's made in the Foxconn plant in Chennai, India. Practically speaking, they don't pay the plant for their labour in US dollars (USD) since the bosses and the workers aren't going to be able to go to the local shops and services and exchange the USD for the product, so they'll be asked to pay in Indian Rupees (INR), which is the national Indian currency. Now, Apple's accountants (god forbid they aren't the same ones Enron used) would have to sit and work out the exchange rate from USD to INR, exchange their USD to INR and then pay the plant the exact amount before they agree to start building their phones. However, the USD to INR exchange rate isn't fixed... it changes. If one day, 1 USD can buy you 80 INR and the next day 1 USD can buy you 90 INR, you get 10 more INR available to you. If you were Apple, you'd be incredibly happy since you use up less USD to pay, say the 8 billion INR bill to make 1 million units of the next iPhone. If the exchange rate falls to 70 INR, then by simple division Apple is going to need to give away more USD to pay the bill and Tim Cook won't be a happy CEO when it comes to the next Q4 shareholders' meeting.

Another reason that's less commonly mentioned in finance is an institution's foreign reserves. Let's take VW for example. They don't just sell their cars in Germany, do they? They sell in every country in the world

(probably). As a result, they're going to set up a company in each country under the name 'VW' to be able to collect money in that country's local currency. Let's take Japan for instance. They'll probably sell thousands of units of the VW Golf, all been paid for by Japanese customers with the Yen (JPY) that they've earned from their employment. That's all good, but what if VW wants to wire a fraction of its JPY revenue to their main account in Wolfsburg to pay some of the bills at the main factory at Wolfsburg as well? Assuming their accountants are not the same people that do their emission tests, they'll have to convert JPY to Euros (EUR) first since if they directly try to wire the JPY reserves from their Japanese company accounts, they'll find that the German company charging the electricity bill isn't going to accept JPY at all (for the exact same reason above). So, they'll first convert their JPY reserves to EUR, transfer it over to their European accounts and then pay off the relevant bills they need using EUR. How intuitive!

Exchange rates

Right, now that we've discussed the purpose of FX, let's go on and discuss how exchange rates are quoted.

We're going to repeat this point again and again, so make a note of this point:

Any trade can be simplified to two features.

What do I mean by this? Well, when we trade a currency, we have a currency that we buy and a currency that we sell. So, there's only two currencies involved in a trade, depending on if we are buying or selling them. For this reason, we will be trading in *currency pairs*.

Take our previous example of Apple having to pay Foxconn chennai 8 billion INR to pay for its manufacture of the iPhone. Taking an exchange rate of 1 USD = 80 INR, it's pretty simple to assess Apple's cash flow by drawing a table:

	USD (\$)	INR (₹)
1. Buy 8 billion INR at an exchange rate of 1 USD = 80 INR	– 100,000,000	+ 8,000,000,000
2. Pay Foxconn a bill of 8 billion INR.		– 8,000,000,000

So, they've lost 100 million USD to convert it to 8 billion INR, which they've then had to transfer to Foxconn in order to pay the bills. This method of drawing a table is incredibly useful to track changes in cash flow and we'll be coming back to this repeatedly over the next lectures.

One way to think about the exchange rate is the ratio of **value** of one currency to another. Here, we've specified that the value of 1 USD is 80 INR – that's how much it costs to be able to *buy* USD by *selling* INR. Unsurprisingly, there is a standardised notation for this sort of thing to save traders time and make sure everyone talks the same way, which is what we're going to talk about.

Currency Pairs

We often write currency pairs using the currency codes and separate them with a backward slash:

USD/INR

This looks pretty obvious... because it is. This is how you write a currency pair. Some of you might be wondering “well... How do I know which currency I'm buying and which I'm selling? Is there an order in the convention that specifies this without needing to be specific?” And you'd be right.

The first currency you specify:

USD/INR

is known as the base currency. This is always given a value of 1. In this currency pair, this ratio refers to 1 USD.

The second currency you specify:

USD/INR

is known as a quote currency. This is not fixed to any value, so it can take any value it wants. Why is this the case?

The base currency is given the name ‘reference element’. This is because the quote must be read as such:

How much you have to pay in the quote currency to be able to buy 1 unit of the base currency.

Here, the USD/INR is a currency pair that specifies how many INR you need to sell in order to buy 1 USD. Take this example:

$$\text{USD/INR} = 80.00$$

This currency pair quote tells me that I need to pay (i.e. “sell”) 80 INR of my own to be able to buy 1 USD. Yes, inherently, the quote should really mean 1.00/80.00 i.e.

USD/INR

 1.00/80.00

but since we've specified that through convention, the base currency is fixed at 1 unit (I've just put 1.00 to keep the decimal points identical but it does literally just mean *1 unit*), we don't need to constantly rewrite it. We can save ourselves the time to just write the units of the quote currency we need to sell to buy 1 unit of the base currency. It's a bit like in maths where we write $f(x) = x^3 + x^2 + 3$. We don't need to write the entire polynomial again: we can just write $f(x) + 3$ or $2f(x)$ to save ourselves time when we mean $x^3 + x^2 + 6$ and $2(x^3 + x^2 + 3)$ respectively.

So, when buying the base currency such as the USD through selling INR, we read the currency pair from right to left.

However, what if we want to do what Apple did and *sell* the USD to get INR?

Some of you might sit there and think “hmm... What if I write INR/USD? Then, I can just do $1.00 / 80.00 = 0.0125$ and then simply write:

$$\text{INR/USD} = 0.0125$$

That seems like the logical step, no?

NO, THAT'S NOT CORRECT. Well, it's sort of possible *but* in the language of FX trading, that's not what happens.

To discuss why this is not correct, we need to discuss another convention: currency rankings.

Currency rankings

To save you from the history lesson that I could give about this, there is a specific order that you write the order of currencies **REGARDLESS OF IF YOU ARE BUYING OR SELLING one of them.**

The order is based off a rankings list of currencies called **the majors**. I like to think of them the same way I think about the ATP Tennis Rankings, or the F1 Driver Championship rankings. There's a convention in tennis that the first name on a tennis match scoreboard is the player that's higher ranked. For instance, if Novak Djokovic (No. 6) faces John Isner (No. 33), you'll see the notation of a game with Djokovic's name first on the scoreboard, followed by Isner's score:

	Set 1	Set 2	Set 3	Game: Semi Final, ATP Masters 1000 Indian Wells, 2014.
Djokovic, SER. (no. 6)	7	6	6	
Isner, USA (no. 33)	5	7 ²	1	

We do the same with currencies: Let's start with the rankings before the notation.

Here's the ranking of currencies for the majors:

1. Euro, EUR (€)
2. Great British Pound Sterling, GBP (£)
3. Australian Dollar, AUD (\$ or A\$)
4. United States Dollar, USD (\$)
5. Canadian Dollar, CAD (\$ or C\$)
6. Swiss Franc, CHF (CHF)
7. Japanese Yen, JPY (¥)

So, in currencies, we write the higher ranked currency first as *the base currency*, followed by the lower ranked currency as the quote currency.

For instance, if we're trading between Euros and Pounds, we'd first write EUR (no. 1) followed by GBP (no. 2):

EUR/GBP

This information does not include whether I'm buying or selling the Euros – it is not required in this currency pair. We purely write the order by precedence of its rankings in its majors.

Let's trade between the Swiss Franc (CHF, no. 6) and the Australian Dollar (AUD, no. 3). The AUD is higher in ranking, so we write:

AUD/CHF

What are exotics and cross currency pairs?

Very good stuff. But wait, there are more than 7 currencies in the world. What about the INR? Or even, what about the Chinese Yuan (CNY)? Russian Rouble (RUB)? These are known as exotics! Why are they called exotics? Essentially, no one really stores money as CNY or RUB unless they're making transactions within those countries e.g. China and Russia respectively. It's more business minded to hold your money as USD or EUR as it's more geopolitically stable and more people can buy or sell USD and EUR with you compared to CNY or RUB. They're *unique* currencies to be holding i.e. they're **exotic**.

So, what do we do with CNY or RUB? Well, since they're not on the majors list, we pair them up with one of the currencies on the majors list and they'll always take the place of the quote currency e.g.

CAD/CNY
JPY/RUB

You're probably starting to get the message now... *the majors always come first*.

Conveniently, the Americans in the US have a special name for currencies that don't involve the USD as either the base or quote currency – **cross currency pairs**. For instance, the AUD/CHF pair is a cross currency pair. Even better, any currency pair that involves the EUR is given the term **euro crosses**. The EUR/GBP is a very common euro cross that you'll encounter given how much it's traded (known as its 'volume') and its importance in economics and politics (hehe, Brexit).

Right, how to sell currency pairs. Let's get back to that for a bit now.
We wanted to sell USD and buy INR. The previous quote we had was:

USD/INR = 80.00

which meant that to buy 1 USD, we had to sell 80 INR. Lovely. You'll remember that this was by convention since USD was the base currency and INR was the quote currency. If you think of it, to buy INR and sell USD, it's literally going the other way. *To buy INR and sell USD, you sell 1 USD and you earn (i.e. buy 80 INR).* **However, we still use the same quote but just read it the other way!** This time, we're selling the base currency at 1 USD and whatever number is quoted is how much of the quote currency (in this case, INR) is what we get!

To make it easier to read currency pairs, I like to think of it as this:

To buy/(sell) 1 USD / INR = you sell/(buy) 80.00 INR

where you would e.g. cross out the (sell) next to USD if you were buying USD and correspondingly cross out the word in brackets (buy) on the other side to read:

To buy ~~(sell)~~ 1 USD / INR = you sell ~~(buy)~~ 80.00 INR

It's as simple as that.

How do you trade two different exotics between each other?

But what if you wanted to trade two currencies that were exotic and **NOT** majors e.g. the CHY and RUB example from earlier? There's a choice of two possible options here:

1. Simply trade the exotic currencies directly. There's no real ranking to compare CHY and RUB but you can use your gut instinct to work out which might be bigger – China's a bigger economy than Russia, so you'd be logical to quote CHY/RUB, taking CHY as the more important base currency.

This might be a problem however when it comes to liquidity. To take a layman's definition, liquidity is a measure of how easy it is to buy or sell something. Cash is very easy to sell... It's cash. You can go into any supermarket and sell your cash for a loaf of bread (that's what we mean by 'selling' cash). A house, however, is illiquid. It's harder to sell a house and get money in return for it to spend on other things.

If lots of people are willing to sell and lots of people are willing to buy, liquidity is high. If either of the two don't exist (or exist in low numbers), then there's low liquidity. Since few people would wish to trade in their CHY directly into RUB, it would be most likely that the liquidity is low and as a result, the bid-ask spread might be ridiculously huge.

2. You might choose to sell one of the exotics and buy one of the majors first as an intermediary currency store and then trade again into the other exotics you wish to buy. What this means is e.g. you might want to first sell CHY and buy USD. Then, you might wish to convert your USD to RUB by selling USD and buying RUB.

This is an option that's definitely more likely to be taken by traders. However, since you're doing two trades rather than 1 (CHY → USD, then USD → RUB compared to CHY → RUB), you might find that you spend twice as much on fees and commission although if the bid-ask spread is tighter, you're less likely to lose money than trading CHY/RUB directly.

Right, now that's exchange rates and currency pairs covered from its fundamentals. What's next? Ah, yes. The types of exchange rates!

The two types of FX exchange markets

So far, we've spoken about why exchange rates exist, how you write them using currency pairs and how you can buy and sell from one currency to another. So far, this is pretty elementary stuff. Now, we're going to go up a level and talk about the different types of exchange rates that exist. What you'll be covering here will also be fundamental in Lecture 4 so I'd suggest you pay good attention to what we cover here in order to make the lecture on derivatives (Lecture 4) much easier!

Currencies are *traded* on the **foreign exchange market**. You'll know this already but just for clarification, a market is an area (either one that exists in the real world or digital) where buyers and sellers can meet to buy and sell goods respectively. On the foreign exchange market, currencies are bought and sold. We've already mentioned two reasons above – the ability to trade internationally outside of your domestic country, being able to invest in different countries that use different currencies. Another reason people trade currencies is the exact reason that people trade stocks... They think a currency will go up **OR DOWN** in value soon and they can profit from the difference in price between now and a point in the future. This third reason is called **speculation** i.e. the idea that they think they'll benefit from a change in price. Often, there is a misconception

that speculation is the idea of prices only going up. This is not the case – prices can go down so you'd benefit from *shorting* an asset (in this case, the asset being a currency).

Right. Now we've cleared that up and out of the way. There are two types of FX exchange markets that you can find inside, er... FX exchange markets:

1. Spot exchange markets
2. Future exchange markets

Let's discuss spot exchange markets first, which is far simpler to understand.

Spot exchange markets

It's pretty simple: they are markets where you can trade currency **right now** i.e. *on the spot*. Often, I find people struggle to understand what a spot market is so I hope that the definition of things being bought and sold 'on the spot' is the easiest to get a handle on before you encounter more complicated ideas later in your financial education. You would buy the currency right now and have it in your hands the same second you pay the asking price for it. It's just like walking into a shop, paying the shopkeeper and walking out with the goods all on one go.

We tend to say (thanks to financial lingo) that you *take immediate delivery of the asset* which actually just means the same thing as walking into the shop and walking out with what you want in your hand all in one go. This doesn't just apply to currencies (as you will find out in Lecture 4) – spot markets apply to pretty much anything you can think of whether that be oil, gas, feedstock etc. We specify in this case that we're discussing spot exchange markets to refer to the idea of currency being traded.

You'll have encountered spot exchange markets in real life when you visit a Bureau de Change. You walk in and exchange your EUR to GBP (or vice versa) and you get the currency you want on the very same day – there's absolutely no need to wait. In financial markets, companies will do much the same process through FX markets to be able to convert currencies at a specified exchange rate right now at this very moment.

Now, I did mention that with spot markets you can exchange currencies and receive the currency you desire at this very second. What if you don't want to exchange currencies now but instead do the exchange in, say... a month's time (far forward in the future) while locking in an exchange *rate* right now? That's what *forward* exchange markets do!

Forward exchange markets

Now, before I get some questions as 'when are we going to learn about forwards and futures', I would like to stress that they will be the topic of Lecture 4. We only have so long each lecture to cover things so we'll briefly cover forward exchange markets from a high-level overview before we discuss them in greater depth later on in a few weeks' time.

In a forward market, we pay today at a specified price but we get our goods (in this case, our new currency) at some point in the future. To do this, we have to first buy a contract called a *futures* contract. Yes, some of you will be aware that two types of contracts exist – forwards and futures. As I said, there are differences between the two and they'll be the subject of lecture 4.

Take again the example of Apple. I briefly pointed out that if the exchange rate for USD/INR falls from 80.00 to 70.00 over the course of, say, 6 months or something, they'd lose out on an extra 10 INR per USD. Of course,

it would be very ideal for Tim Cook and his accountants to lock-in today's exchange rate of 80.00 and use that to pay off Foxconn in 6 months time if that's the payment structure that they've planned instead of the 70.00 INR per USD as they would have to cough up an extra 0.125 USD to make up the difference. So, Apple buys a futures contract for a small fee to have the currency rate fixed for one day in the future at 80.00 INR if they expect that it'll fall in half a year to 70.00 INR. It's certainly a risky strategy as if the USD/INR goes up to 90.00 (which we often say as '*the USD getting stronger against the rupee*') then they've lost a small amount of money paying the fee to secure the futures contract but it's worth it if they feel their forecasting measures are accurate and will pay out (i.e. be profitable to them) in the near future.

That's all that I am going to say about forward exchange markets. Oscar and I in lecture 4 will explore this area with much more depth so make sure you understand this point well enough as we'll discuss it again soon.

Exchange rate systems

Himanish earlier introduced the simple yet fundamental concept of supply and demand economics. You can draw a line for both the supply and demand for a good by plotting the price the good is valued at against the amount of the good that people are prepared to pay (i.e. the demand). Just to be clear, a good (noun, n.) is just an item of any description, whether that be a loaf of bread, a barrel of oil etc. Cash is a type of good as well! You can use it in exchange for other goods when you complete a trade with a seller (not so bad, huh?).

So, you could ask "can we draw supply and demand curves for currencies?" Yes, we can. When we draw supply and demand curves for currencies, the axes we use actually have specific meanings this time compared to the usual 'price' and 'quantity' that you saw earlier. Take the EUR/GBP currency pair. Our x-axis this time is 'quantity in EUR, € millions (m) per day' (since currencies are traded on a daily basis) and the y-axis is 'price, GBP £ per EUR €'. If you are a bit confused, remember we are fixing the unit of EUR to 1 and we're looking at how many GBP we need to pay for 1 EUR.

Floating exchange rate systems

Supply and demand of currencies

Supply curves

For completion, let's again summarise the supply and demand curves separately. First, we'll work out the demand curve. If the EUR/GBP is priced high i.e. we need to pay a high amount of GBP for 1 EUR, very few people are going to want to buy EUR, hence the *quantity of EUR demanded* is going to be low. However, if the EUR/GBP is priced low, a lot of people are going to want it as it's affordable and cheap hence the *quantity of EUR demanded* is going to be high. For this, we're going to draw a straight line that slopes downwards:



FFS, it's £ per €. Needs changing – the x-axis is fine.

Demand curves

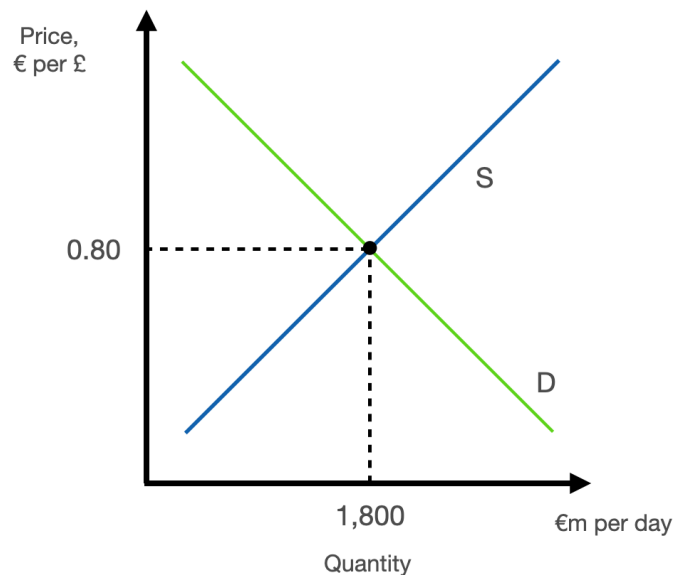
Now, let's think about the sellers. At a low price, very few sellers are going to be prepared to *supply* the market with EUR as they think the price that they have to sell at (based on the market price for EUR/GBP) is too low and hence they won't make a reasonable margin or profit. However, if the price of the EUR/GBP is much higher, they'll be more incentivised to pull their reserves of EUR and supply it on the market for people to buy. For this reason, at low EUR/USD, the *quantity of EUR supplied* will be low but at higher EUR/USD, the *quantity of EUR supplied* will be high.

Let's draw that separately on another set of axes.



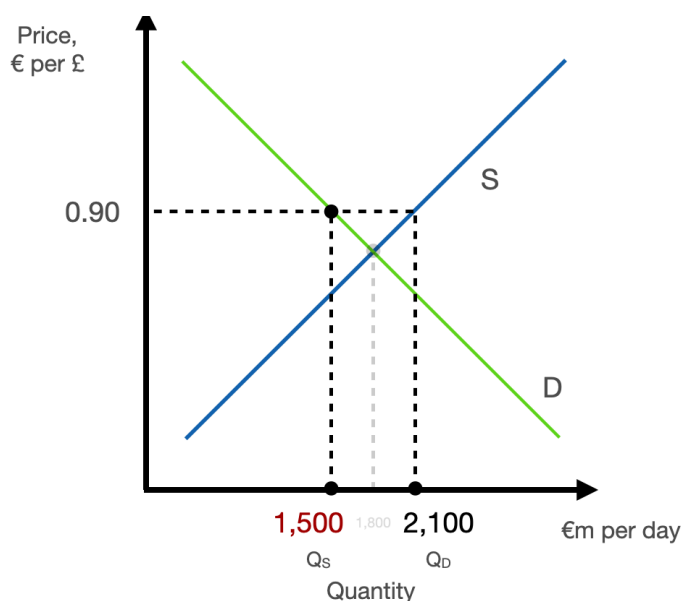
Combining supply and demand curves together

Now, this is interesting. What if we go one step further... what if we draw both ones, one on top of each other (assuming both axes are the same scale)? What would happen here?



We get a complete supply-demand curve for the EUR/GBP! Notice that since we have two *straight* lines here, they're going to intersect at one given point. This point is critical (as Himanish has mentioned) as it's what we know to be the **equilibrium point**. Think of the equilibrium point as the point on the graph where the flow of supply and demand are equal i.e. in equilibrium (just like forces in mechanics or one-step reactions in chemistry). In the context of macroeconomics here, it means that the EUR is being sold as quickly as it's being supplied into the market. If the price of the EUR/USD is set to £0.80, a seller would be guaranteed a sale as they would be supplying €1,800 million per day and finding it being sold all by the close of business.

If the price by sellers was artificially set higher e.g. to £0.90, there would be a difference between the quantity supplied and the quantity demanded so the sellers would find some EUR left over at the end of the day since there aren't enough buyers interested in purchasing EUR at £0.90 but instead at lower prices (as you can see on the demand curve). At £0.90 per EUR, there's only demand to buy €1,500 million but suppliers would be supplying €2,100 million, leaving €600 million unsold – a seller's nightmare. For this reason, they want to make sure they're as close to the equilibrium price possible to ensure all of their supply of EUR is sold out.



Here, the exchange rate quite obviously (in the graph on the previous page) was $\text{EUR/GBP} = 0.80$. This type of exchange rate has a special name: **floating exchange rate system**. I like to think of it as a boat that's sitting on water with a tidal wave passing by. The boat will move up and down in sync with the passing waves while it stays afloat, without any external interference. The reason I mention 'external interference' is that the exchange rate of a currency pair is not influenced by any country's government – it is left purely to the market's 'free market forces' of demand and supply to reach some sort of an equilibrium point at, say, 0.80 GBP and everyone trades the currency pair at that price.

The issue of floating exchange rate mechanisms

However, if you are Boris Johnson (or Rishi Sunak, or even Liz Truss depending on who becomes Prime Minister), having your own country's currency priced quite low is a bit of an issue. Of course, it looks terrible on economic policy since the rest of the world can buy your 1 GBP at 1.25 EUR (that's 1 EUR / 0.8 GBP, for those wondering) meaning that the pound is low in value, but it also means that for factories and industries based in Britain, the price of imports are quite high. Having to pay £0.80 for €1 can be quite steep so industries might be enticed to increase their imports if, say, the pound fell against the euro to around £0.50 per €1. That way, you can cough up fewer GBP to pay for something in EUR. Economics is a balancing act in essence based on the government policy that's being implemented any one time: should the GBP be priced higher e.g. $\text{EUR/GBP} = 1.50$, or should the GBP be cheaper so imports are easier to buy e.g. $\text{EUR/GBP} = 0.60$.

So, how can countries intervene to adjust their country's exchange rate? For that, there are several other classes of exchange rate systems, which we'll get onto next – after a discussion on purchasing power parity!

Purchasing power parity

I'll make a confession here: I do really (and I'm being truthfully honest) hate this term... 'purchasing power parity'. It's not really the term itself that I dislike but the way it's explained in modern economic textbooks. I'm going to try and develop this idea from the ground up although if you try reading a conventional economics textbook, you'll find some (quite honestly) woeful descriptions. Anyhow, let's crack on.

Let's start with a unit of currency. I've mentioned it before but I'll mention it again: a unit is just another way of saying 'one' i.e. 1. A unit cube is a cube that measures 1 cm³, a unit circle is a circle that has a radius of 1 (metre, centimetre... whatever). A unit of currency (often shortened to 'unit currency') is just a single value of the currency e.g. 1 GBP, 1 EUR, 1 USD, 1 JPY... you get the idea.

Think about the GBP/USD currency pair. Right now (08/06/22), it's valued at 1.21 – that's 1.21 USD for every 1 GBP. That's an exchange rate that you find in any exchange *market*, being traded between different institutions to buy and sell GBP and USD respectively.

Purchasing power is basically how much you can buy for 1 unit of currency. You might have heard back in the old days that a Freddo would cost around 10p in 2000 although it's now around 25p in 2022. I should stop here for a moment and explain for the uninitiated amongst you that Freddos are a classic Cadbury's chocolate that was known for being very cheap years ago although due to inflation and rising costs, it's used as a measure for the increasing living cost in the UK. Conveniently, there even exists a humorous 'Freddo index' in its name to track its price over the last 20+ years. Back to the main point. Well, in 2000, you could walk into a shop with £1 and buy 10 Freddos (100p / 10p). 20 years later, you could only buy 4 Freddos (100p / 25p). That means that the purchasing power of a unit GBP with respect to freddos has gone down in the last 20 years i.e. the amount of freddos that you can get with a single coin has gone down over time.

Now, think of this: you do your weekly shopping at Sainsbury's which includes food, drink, meat (unless you don't eat meat), washing-up liquid, toiletries etc. and that comes to around £30. Now, imagine walking into a Walmart store in America and picking up the exact same list of groceries in your list (and I mean, literally the exact same) and it comes to around \$45. That means that by some simple maths, £1 in England (in theory) buys you \$1.50 worth of goods in the USA. That's an *actual* exchange rate for GBP/USD of 1.50.

What's going on here? Why is there an exchange rate for 1.21 in the exchange markets for GBP/USD but 1.50 when we actually compare the same sample shopping list? This is because the exchange markets reflect what we call '*short-term speculations*' whereas the grocery lists actually reflect longer-term effects of inflation that exist in both the UK and USA economies. Exchange market exchange rates are called **nominal exchange rates** as they're just determined by the equilibrium price from buyers and sellers in a market – 'nominal' means *only by name*, which means the value that's quoted is unrealistic and doesn't reflect the actual nature of the exchange rate in reality. In reality, the GBP/USD exchange rate is much higher at 1.50 if we look at a specific *basket of goods*, which we call a **real exchange rate**. There's actually a theory which states that changes in exchange rates for a currency pair in the long-term are due to the differences in inflation rates between the two countries – this is conveniently called the **purchasing power parity of exchange rates**. The term 'parity' is a reference to the equality between how much you can buy when you compare the USA to the UK, which comes from the idea of how 'equal' one thing is to another. We're not comparing currencies based on their exchange rate, but instead on how much you can buy in one country compared to another.

Another way to think about purchasing power parity is how much you can buy for a specific wage. For example, I (Ahmad) am German and so I work in Berlin but Himanish here works in London. My weekly wage is around €110 but Himanish earns £80 per week. Who *earns* more? This is a bit of a hard question to answer. Yes, we could go do the exchange rate conversion and we'd find out that Himanish earns the equivalent of €100, which would mean that my weekly wage of €110 is higher. However, if Himanish can go grocery shopping and buy enough supplies at Asda that would last him 14 days on his £80 per week wage but I can only buy enough supplies at Lidl for 10 days, Himanish would be the one who's earning more *because of the lower effect of inflation in the UK compared to Germany* – this is, of course, purely hypothetical and doesn't reflect the current situation (probably). That's the purchasing parity power at play, where Himanish has more purchasing power than I do, even though I earn more Euros than he does.

In reality, the price of a shopping list should be identical wherever you go. This is rarely the case due to long-term inflation rates being different in both the USA and the UK. However, that works differently to what happens in the foreign exchange markets, where the prices are determined by the supply and demand between sellers and buyers respectively.

You won't come across purchasing power parity too often but it's an important topic in macroeconomics that we thought would be useful to cover. Right, onto other types of exchange rate systems.

Other types of exchange rate systems

Fixed exchange rate systems

Sometimes, a government might wish to be directly in charge of making sure an exchange rate that involves its own currency is *fixed* at a specific value. These types of exchange rate systems are **fixed exchange rate** systems. Just to be clear here, this **DOES NOT** mean that it'll always stay precisely at one specific number. As you'll see momentarily, this used to be the case up until around the 1970s when a certain Richard Nixon did something that changed economies and finance forever.

Gold standard

One example of a fixed exchange rate system was the **gold standard**. For hundreds of years, any currency whether that be the pound or the dollar was linked to a specific weight of gold in the country's central bank. This meant that e.g. if you were in America, you would be able to walk into the Federal Reserve and be able to exchange a \$10 note for nearly 15 grams of gold from the 1870s to 1914, and again from the early 1920s to 1932. In England, the same 15 grams of gold could be exchanged for £2.06, which implied an exchange rate of GBP/USD $\approx$ \$4.87. The same could be done in France, Germany, England as well. However, there was a problem. Since there was a direct link between the weight of gold that a central bank stored and the amount of paper money it could issue, it was hard for England and other countries to issue more paper notes in order for the Government to pay for services and other stuff such as imports – the exchange rates were fixed and so the rates themselves determined the amount of paper currency in the country's money supply. One way to solve this was to mine for more gold but that would take time, be a waste of effort for the little reward and also be difficult to find areas of land that have gold (unless you could invade, say, another country... but no one really did that – much, anyway).

Bretton Woods

This whole gold standard business changed *slightly* after World War 1, the intermittent Great Depression and World War 2. In 1944, another system came about called the Bretton Woods system. Essentially, 44 countries sent delegates to, er... Bretton Woods in New Hampshire, USA. A new financial system was needed that was less rigid than the old gold standard but still standardised enough so it was under control, without it falling apart. The Americans at this point had managed to hoard over 66% of the world's gold reserve from travelling through other countries across Europe, so it decided to be the main country in charge of a new type of gold standard where the USD would be linked to its reserves of gold, and in turn all other countries including England, France, Australia etc. would fix their currencies to the dollar at a near-fixed exchange rate, with some very small fluctuations allowed. This was a huge win for the USA as it would essentially be in charge of the world's economies through the dollar by manipulating its own gold reserves. Unsurprisingly, just as many American policies have shown post-WW2, the Bretton Woods system didn't work. Most countries were facing their own economic problems and decided to take back some of the gold that belonged in the Federal reserve, most notably France who even sent a battleship to transport their gold back from the USA.

Fiat currency

As a result, Richard Nixon (USA President) found a battle of trying to print more money in the face of his country's stagflation but also facing dwindling supplies of gold. He fiddled with the gold to US exchange rate from 1971 up until 1973 when he decided to decouple the dollar from the gold. Now, the dollar and every other currency in the world was essentially 'backed' by nothing and the dollar was 'floated' so it couldn't be exchanged for gold anymore. No one could exchange it for a specific value of gold anymore that was guaranteed by a central bank. People now believe that a dollar note gives you the right to purchase something for a dollar based on what it says on paper. This is what we know now to be a **fiat** currency.

Nowadays, a few countries fix their exchanges to another currency, namely the dollar. To keep the exchange rate fixed, the country's central bank dabbles in buying or selling its own currency in order to keep the e.g. USD/CNY currency pair as constant as possible, if we were talking about the Bank of China. *Just to be clear, China does **NOT** have a fixed rate exchange system with the USD – we just used it here as an example.* If the USD/CNY moved from the intended fixed rate of 6.50 to 7.00, it would imply that the CNY weakened against the USD as 1 USD gives you more CNY, so the Chinese government would buy CNY in foreign exchange markets to lower its supply and shift the exchange rate back to 6.50, which is known as making the CNY '*stronger against the dollar*'. Likewise, if the USD/CNY fell to 6.00, it would mean that the USD became weaker against the CNY (or the CNY got stronger against the USD – notice the opposite effect on the other

side), so the Chinese government would sell some of its CNY reserves to restore the price back upwards to 6.50 by increasing the supply. The overriding idea is this: the government of the country in question wants to make sure its currency is fixed as much as possible at a specific value (or as close as possible) by selling or buying its own currency to affect its supply in the foreign exchange market.

Managed exchange rate systems

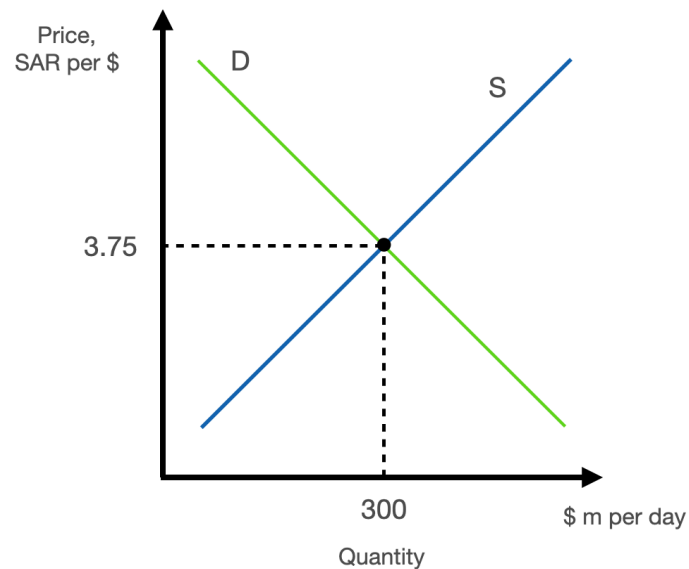
Realistically, governments have other jobs to do than to sit down all day and look at an exchange rate e.g. establish policy, liaise with trade partners, get exposed by an undercover journalist in the newspapers – you know, the usual stuff (just kidding). An alternative sort of exchange rate system is a **managed** exchange rate system. They're also called **hybrid systems**. Why is that? Well, in these sorts of systems, there's two mechanisms at play. The first is that the market is free to set the price through the forces of demand and supply. However, sometimes the government might want to step in and sort things out directly to push the exchange rate either up or down in price through selling its central bank's own gold and foreign currency reserves. An alternative way at the government's disposal that we won't discuss this lecture but is also important is through an indirect means of fiddling with the interest rate – either increasing or decreasing it until the right effect is achieved (a bit like right now with the Bank of England going from 0.1% to 1.75% in less than a year.) Managed exchange rate systems can be broken down into different types, although be careful as they are **NOT** fixed exchange systems (as you will see momentarily).

Adjustable peg systems

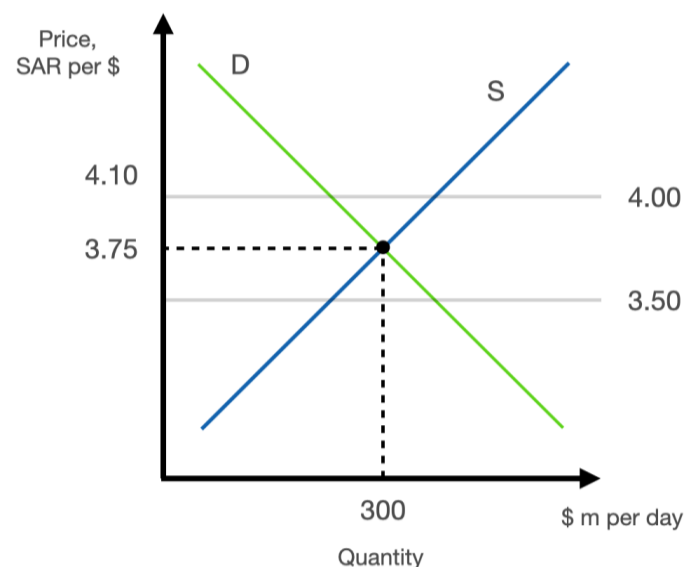
An alternative way of saying that a currency rate is 'fixed' is to say it's 'pegged', a bit like when you put your laundry on a washing line – you fix it on the line to make sure it doesn't move when the wind blows through it. Another term that's used is "anchored" but I think it's easy to visualise why that term is also appropriate. So, we got the idea of a *peg* out of the way. What's an adjustable peg? In fact, if you think back to the washing line analogy, this idea isn't too hard to understand. In much the same way that you can slide an item of laundry up and down the washing line until you find a point that's just right, currency rates can also be moved up and down in value until the central bank feels it's just at the right price.

Normally, we peg two currencies e.g. the USD and the Saudi Arabian Rial (SAR) against each other so that they stay fixed in value *in the short term*. In the long term however, the Saudi central bank might wish to adjust the value of the currency i.e. 'move the peg' up or down in response to economic activity such as an oncoming recession or to spark a growth phase in the Saudi economy. The Bretton Woods system was a good example of an adjustable peg as all currencies were pegged against the USD in the short term. In the long term however, countries would take measures to make sure their daily exchange rate with the USD fell within a certain narrow, pre-established limit. If the price of its currency rose, it would take measures to sell its currency in FX markets to increase supply and decrease its price, but also buy the currency if the price of the currency went down with respect to USD.

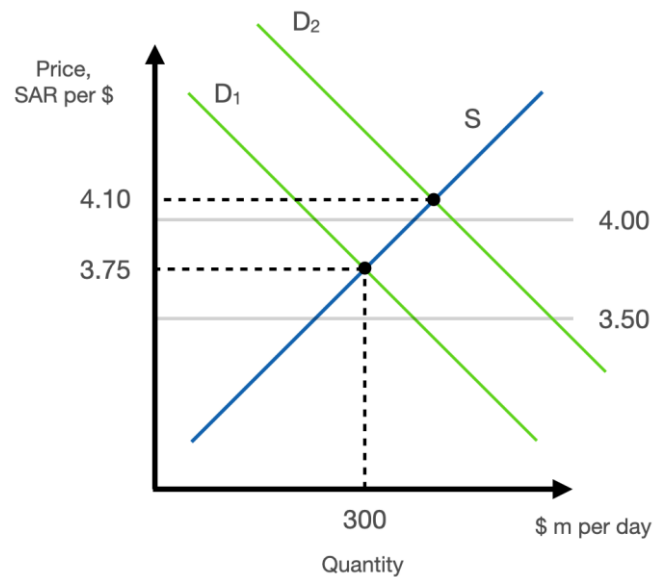
You can actually represent this using an extension of the supply and demand curves that you're familiar with from earlier. Let's first draw our usual supply and demand curves, noting the equilibrium price. We'll draw some pre-established limits in red so that we can show where we expect the equilibrium price to hover around. We assume that demand and supply are once again linear and draw the usual axes that we mentioned earlier for the USD/SAR currency pair.



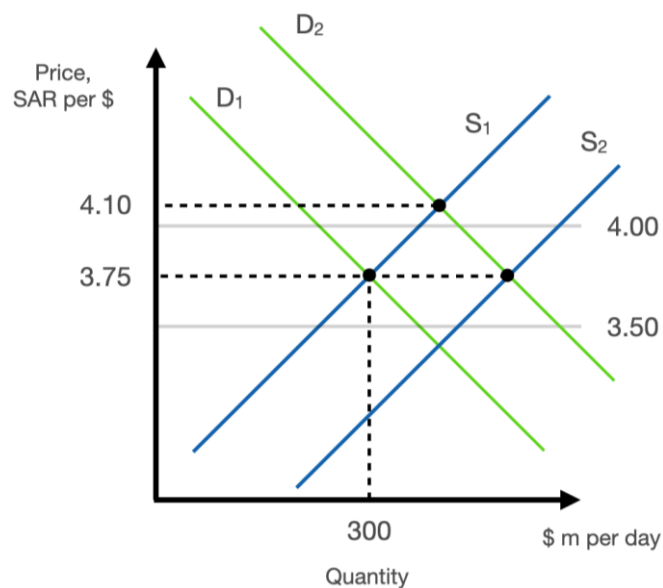
This is fairly intuitive at this point: the exchange rate for USD/SAR is 3.75 SAR and 300 million SAR is converted into USD every day. Let's add some strict limits so that the USD/SAR remains broadly around 3.75 most of the time:



Here, we've specified an upper limit of 4.00 and a lower limit of 3.50 for the USD/SAR currency pair. What then happens if demand for the USD increases? Well, the exchange rate goes up. We can display this by shifting the demand curve along the supply curve (i.e. parallel to the curve S). We assume the supply of SAR is constant so we only observe the price change due to demand. Our new demand curve is called D_2 and our old one is now renamed as D_1 , as you see below:



Oh dear, the Saudi central bank isn't going to like this. The price of the USD/SAR has gone up to 4.10 meaning that the SAR has lost its value as you can buy more SAR per unit of USD. How is the Saudi central bank going to fix this? They want it to come back down to between 4.00 and 3.50 and preferably around 3.75. Hmm, what if they *sell* some of their own reserves of USD so the USD supply increases and SAR supply decreases? Now, the Supply curve S will increase from S_1 to S_2 and a new equilibrium price will be set that's lower than 4.10 but in-between the 4.00 and 3.50 limits:



And now we're back where we started... lovely stuff.

Crawling peg system

To think about a crawling peg, let's think first about two types of systems we know so far. We know already that an adjustable peg systems allows the price of a currency to fluctuate between two hard limits we set artificially such that if the price falls outside of the limits specified, the central bank will undergo some buying or selling of its own currency in order to bring it back within the tolerance we want. Also, we know that in a floating exchange rate system, we simply let the free market supply and demand forces dictate the price of the currency itself.

A crawling peg system is a bit of a hybrid between an adjustable peg system and a floating exchange rate system. It still is a peg system (like we mentioned above) *but* we move the limits slowly to allow the central bank to buy or sell the currency over time in response to economic sentiment. If the price of the currency is increasing over the last 3 months, we move the limits up by a small amount (hence the 'crawl') and establish a new set of limits that the currency has to live in so that it reflects the recent market and economic momentum. Afterwards, the central bank then sells the currency in question so it fluctuates around a new, lower price.

China actually does this with the Chinese Yuan (also known as the 'Renminbi'), where the government pegs the Yuan to different currencies but also can get involved in buying and selling the Yuan itself by first moving the limits up or down respectively and then completing the necessary fiscal policies of buying and selling its currency to meet its objectives.

Managed float

Remember what a 'float' meant in FX? It simply just meant letting the currency price be dictated by supply and demand in the free market, a bit like a boat on the sea with the tides going past to rock it up and down. A **managed float** is similar in FX – it has all the features of a floating exchange rate system but with one specific difference... the government can intervene and buy or sell the currency in order to change its price!

This actually isn't too different to the crawling peg, but instead the price isn't fixed by the government and the central bank in question yet instead it's left to the free market. This is generally the most common type of exchange rate system you'll find in financial markets, with the USD and GBP being prime examples of a 'managed float'. Often, you might find it called a **dirty float** as it's not exactly a *pure* floating exchange system – it has features of it being fixed by central banks to suit its economic policies when it needs to, although its interventions are few and far between (by that, I mean it rarely happens).

Analysing a managed float is much the same as a crawling peg, but just remember that the equilibrium price isn't as harshly fixed by the central bank – it has a lot more freedom to move around to its liking.